1.

```
$ peterson
Process 0 ENTER CS, counter = 1
Process 0 EXIT CS
Process 1 ENTER CS, counter = 2
Process 1 EXIT CS
Process 0 ENTER CS, counter = 3
Process 0 EXIT CS
Process 1 ENTER CS, counter = 4
Process 1 EXIT CS
Process 0 ENTER CS, counter = 5
Process 0 EXIT CS
Process 1 ENTER CS, counter = 6
Process 1 EXIT CS
Process 0 ENTER CS, counter = 7
Process 0 EXIT CS
Process 1 ENTER CS, counter = 8
Process 1 EXIT CS
Process 0 ENTER CS, counter = 9
Process 0 EXIT CS
Process 1 ENTER CS, counter = 10
Process 1 EXIT CS
Process 0 ENTER CS, counter = 11
Process 0 EXIT CS
Process 1 ENTER CS, counter = 12
Process 1 EXIT CS
Process 0 ENTER CS, counter = 13
Process 0 EXIT CS
Process 1 ENTER CS, counter = 14
Process 1 EXIT CS
Process 0 ENTER CS, counter = 15
Process 0 EXIT CS
Process 1 ENTER CS, counter = 16
Process 1 EXIT CS
Process 0 ENTER CS, counter = 17
Process 0 EXIT CS
Process 1 ENTER CS, counter = 18
Process 1 EXIT CS
Process 0 ENTER CS, counter = 19
Process 0 EXIT CS
Process 1 ENTER CS, counter = 20
Process 1 EXIT CS
Final counter = 20
```

2.

```
$ prodcons
Producer: produced 1 (buffer count = 1)
Producer: produced 2 (buffer count = 2)
Producer: produced 3 (buffer count = 3)
Producer: produced 4 (buffer count = 4)
Producer: produced 5 (buffer count = 5)
Producer: buffer fCuonsumer: consumed 1 (buffer count = ll, waiting...
4)
Consumer: consumed 2 (buffer count = 3)
Consumer: consumed 3 (buffer count = 2)
Consumer: consumed 4 (buffer count = 1)
Consumer: consumed 5 (buffer count = 0)
Consumer: buffPrer empty, waiting...
oducer: produced 6 (buffer count = 1)
Producer: produced 7 (buffer count = 2)
Producer: produced 8 (buffer count = 3)
Producer: produced 9 (buffer count = 4)
Producer: produced 10 (buffer count = 5)
Producer: buConsumer: ffer full, waiting...
consumed 6 (buffer count = 4)
Consumer: consumed 7 (buffer count = 3)
Consumer: consumed 8 (buffer count = 2)
Producer: produced 11 (buffer count = 3)
Producer: produced 12 (buffer count = 4)
Producer: produced 13 (buffer count = 5)
ProduConsumer: cceonrsumed 9 (b: ufbfuerf fcer full,ount  = 4)
waCiotning.s.umer: c.o
nsumed 10 (buffer count = 3)
Consumer: consumed 11 (buffer count = 2)
Consumer: consumed 12 (buffer count = 1)
Consumer: consumed 13 (buffer count = 0)
ConsumerP: buffer empty, waitiroducer: producng..e.
d 14 (buffer count = 1)
Consumer: consumed 14 (buffer count = 0)
CPonrsumeodur: buceffre:r  proedmpty, wauiting..ced 15 (buffer .count
Producer
: produced 16 (buffer count = 2)
Producer: produced 17 (buffer count = 3)
Producer: produced 18 (buffer count = 4)
Producer: produced 19 (buffer count = 5)
Consumer: consumed 15 (buffer count = 4)
Consumer: consumed 16 (buffer count = 3)
Consumer: consumed 17 (buffer count = 2)
Consumer: consumed 18 (buffer count = 1)
Producer: produced 20 (buffer count = 2)
Consumer: consumed 19 (buffer count = 1)
Consumer: consumed 20 (buffer count = 0)
Producer-Consumer completed successfully
```

3.

```
$ readwrite
Reader 2 ENTER, shared_data = 0, readers = 1
Reader 2 EXIT
Reader 3 ENTER, shared_data = 0, readers = 1
Reader 3 EXIT
Writer 1 ENTER, shared_data = 1
Writer 1 EXIT
Writer 2 ENTER, shared_data = 2
Writer 2 EXIT
Writer 2 ENTER, shared_data = 3
Writer 2 EXIT
Writer 2 ENTER, shared_data = 4
Writer 2 EXIT
Writer 2 ENTER, shared_data = 5
Writer 2 EXIT
Writer 2 ENTER, shared_data = 6
Writer 2 EXIT
Reader 1 ENTER, shared_data = 6, readers = 1
Reader 1 EXIT
Reader 2 ENTER, shared_data = 6, readers = 1
Reader 2 EXIT
Reader 3 ENTER, shared_data = 6, readers = 1
Reader 3 EXIT
Writer 1 ENTER, shared_data = 7
Writer 1 EXIT
Reader 1 ENTER, shared_data = 7, readers = 1
Reader 1 EXIT
Reader 2 ENTER, shared_data = 7, readers = 1
Reader 3 ENTER, shared_data = 7, readers = 2
Reader 3 EXIT
Reader 2 EXIT
Writer 1 ENTER, shared_data = 8
Writer 1 EXIT
Reader 1 ENTER, shared_data = 8, readers = 1
Reader 1 EXIT
Writer 1 ENTER, shared_data = 9
Writer 1 EXIT
Writer 1 ENTER, shared_data = 10
Writer 1 EXIT
Readers-Writers completed successfully
```

4.

```
$ dining
PhilosophePhiloPr s0o: THINKING
Philosopher p0he:rP hiloso HhUilosNGRY
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PhipHIhlehrer NKING
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ckr ed fork2 1:
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Philosopher 3: picked fork 4
Philosopher 3: EATING
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NKING
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Philosopher 0: THINKING
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fiGnisPhhpRhYe
r 2i: pPhloPhilsiiolphosopher 1: pickeer d0:cked fork 3
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P: pichikelosopher 2: THINKINildo Ghf
osorpher 3: THINKING
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Philos3: HUNGRr 4: HY
Phok 3UNGRY
pher 0
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osopher 2: EATING
ilosopPher 3: picked fohirk 3
Plosopherhilosopher 3: picked fork 4
P 2: finished eating
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ked fork P1
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2: EATINGPhilosopher 0: THINKING

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Philosopher 1: THINKING
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Philosopher aTl3loH : EATINGINKING
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P picked forohskoi 2
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Philosopher 3: THINKING
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Philosopher 3: picked fork r3
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 fork 0
Philosopher 4: picked fork 4
Philosopher 4: EATING
Philosopher 4: finished eating
Philosopher 4: THINKINGPhilos
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oPhpiloher 4s:opher 0: picked fo TrkHI N1KI
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TING
Philosopher 0: finished eating
PPhhilosopher 0: ilosopher 4: picked fork 0
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d forkP 4
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osopher 4:
HUNGRPPlh ilosopher5 0 c: TyHhilosocINpYhK
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compl 4e:t picked ealdl 5 cycles
 fork 0
Philosopher 4: picked fork 4
Philosopher 4: EATING
Philosopher 4: finished eating
Philosopher 4: THINKINGPhilos
ophePhri los0: picked fork 0
oPhpiloher 4s:opher 0: picked fo TrkHI N1KI
PhilosoNpherG
 0:Phil Eosopher 4: AHUNGRY
TING
Philosopher 0: finished eating
PPhhilosopher 0: ilosopher 4: picked fork 0
PhilosTHoINpher 4: pickeKING
d forkP 4
hPilosohilopsheopher 4: EATING
Philosopher 4: finished eating
Philosopher 4: THINKING
r 0: completed alPl h5i cycles
losopher 4: THINKING
Philosopher 4: HUNGRY
Philosopher 4: picked fork 0
Philosopher 4: picked fork 4
Philosopher 4: EATING
Philosopher 4: finished eating
Philosopher 4: THINKING
Philosopher 4: completed all 5 cycles
Dining Philosophers completed successfully
Philosopher 0 completed 5 cycles
Philosopher 1 completed 5 cycles
Philosopher 2 completed 5 cycles
Philosopher 3 completed 5 cycles
Philosopher 4 completed 5 cycles
$
```